View Article Online / Journal Homepage / Table of Contents for this issue



# Tin Chloride Catalysed Oxidation of Acetone with Hydrogen Peroxide to Tetrameric Acetone Peroxide†

*J. Chem. Research (S)*, 1999, 288–289†

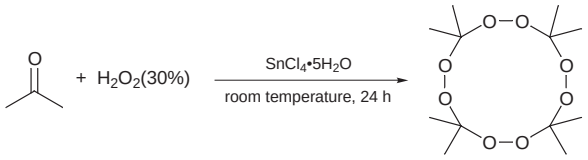
**Heng Jiang,* Gang Chu, Hong Gong and Qingdong Qiao**

*Department of Applied Chemistry, Fushun Petroleum Institute, Fushun 113001, P. R. China*

$SnCl_4 \cdot 5H_2O$ and $SnCl_2 \cdot 2H_2O$ are efficient catalysts for the oxidation of acetone with 30% hydrogen peroxide at room temperature and produce tetrameric acetone peroxide which is identified by molecular weight determination, elemental analysis, FTIR, NMR and MS.

Published on 01 January 1999 on http://pubs.rsc.org | doi:10.1039/A809955C
Downloaded on 17 March 2013

Hydrogen peroxide is one of the most attractive oxidation reagents because it is inexpensive and gives only water as a byproduct. In fact, an enormous amount of work has been carried out on the oxidation of alkanes, alkylaromatics and arenes by $H_2O_2$ in the presence of metal compounds.[1–5] However, there are few reports of metal catalysed oxidation of acetone with hydrogen peroxide. It was reported that cycloalkanones can be oxidized by hydrogen peroxide to cyclic peroxides in the presence of inorganic acids.[6–8] During the investigation of selective oxidation of olefins catalysed by metal compounds with hydrogen peroxide, we found that occasionally an unknown organic crystalline substance is obtained when acetone is used as cosolvent. Detailed research results show that $SnCl_4 \cdot 5H_2O$ can catalyse the oxidation of acetone with 30% hydrogen peroxide (Scheme 1).




**Scheme 1**

Dimeric and trimeric acetone peroxide had been thoroughly studied many years ago.[9] For example, dimeric acetone peroxide was obtained by treating acetone with solid $H_2SO_5$ in diethyl ether while trimeric acetone peroxide was prepared by reacting acetone with aqueous hydrogen peroxide in the absence or presence of concentrated HCl. Another method for the preparation of trimeric acetone peroxide is by autoxidation of diisopropyl ether with molecular oxygen.[10] It can be seen from Table 1 that there is no reaction upon treating acetone with 30% hydrogen peroxide for 24 h. Upon the addition of a catalytic amount of $SnCl_4 \cdot 5H_2O$ or $SnCl_2 \cdot 2H_2O$ to the reaction system, a large amount of white solid is produced. It is found that this white solid is tetrameric acetone peroxide. The catalytic activity of $SnCl_2 \cdot 2H_2O$ is lower than that of $SnCl_4 \cdot 5H_2O$ under the same reaction conditions. Table 1 also shows that the yield of tetrameric acetone peroxide increases with an increasing amount of $SnCl_4 \cdot 5H_2O$. In order to exclude the possibility of oxidation of acetone catalysed by HCl generated from the hydrolysis of $SnCl_4$, 1 mmol HCl was added to the reaction system. It is found that the yield of tetrameric acetone peroxide was very low under such reaction conditions.

**Table 1** Oxidation of acetone with 30% hydrogen peroxide[a]

| Catalyst | Amount of catalyst/mmol | Acetone/mol | Tetrameric acetone peroxide yield(%) |
|---|---|---|---|
| — | 0 | 0.27 | 0 |
| $SnCl_4 \cdot 5H_2O$ | 0.5 | 0.27 | 19.5 |
| $SnCl_4 \cdot 5H_2O$ | 1 | 0.27 | 36.8 |
| $SnCl_4 \cdot 5H_2O$ | 2 | 0.27 | 41.2 |
| $SnCl_4 \cdot 5H_2O$ | 1 | 0.41 | 15.9 |
| $SnCl_4 \cdot 5H_2O$ | 1 | 0.54 | 13.9 |
| $SnCl_4 \cdot 5H_2O$ | 1 | 0.14 | 44.7 |
| $SnCl_4 \cdot 5H_2O$ | 1 | 0.27 | 19.4 |
| HCl | 1 | 0.27 | 8.9 |

[a] *Reaction conditions*: 30% $H_2O_2$, 20 ml; room temperature; 24 h. The yield of tetrameric acetone peroxide is the isolated yield based on acetone.

In the presence of chelating ligands (EDTA, BIPY) or radical inhibitor (hydroquinone, 2,6-di-*tert*-butyl-4-methylphenol) the yield of tetrameric acetone peroxide is 40.0–41.2%. These results show that the reaction is not *via* radical autoxidation or a coordination catalytic process. Thus, the oxidation of acetone with hydrogen peroxide catalysed by tin chloride appears to proceed *via* a Lewis acid-catalysed reaction.

Generally speaking, organic peroxides are dangerous because they readily decompose. Most organic peroxides such as *tert*-butyl hydroperoxide decompose rapidly in the presence of high valent metal oxides or complexes.[11] Under certain circumstances, tetrameric acetone peroxide is very stable and can be heated to 120 °C for 4 h or refluxed for 120 h in organic solvents such as benzene, chloroform and light petroleum. One of the salient features of tetrameric acetone peroxide is its stability in refluxing organic solvent in the presence of $V_2O_5$, $Co(acac)_3$ or $Fe(acac)_3$. Although relatively safe to handle, tetrameric acetone peroxide, like almost all substances containing peroxidic bonds, should be handled cautiously; it is very explosive if it is struck firmly or fired.

In summary, we have shown that $SnCl_4 \cdot 5H_2O$ and $SnCl_2 \cdot 2H_2O$ are efficient catalysts for the oxidation of acetone with hydrogen peroxide to tetrameric acetone peroxide at room temperature.

## Experimental

30% $H_2O_2$ aqueous solution is a guaranteed reagent, while acetone and $SnCl_4$ are analytical grade. The reagents were obtained from commercial sources and were used without further purification.

*General Procedure for the Oxidation of Acetone.*—Acetone was added to a solution of $SnCl_4 \cdot 5H_2O$ in 20 ml 30% hydrogen peroxide and the mixture was vigorously stirred with a magnetic stirrer at room temperature for 24 h. A white solid was produced gradually. After the reaction was stopped, the unreacted acetone was evaporated on rotary evaporator at 25–30 °C, then the solid was filtered

* To receive any correspondence.

† This is a **Short Paper** as defined in the Instructions for Authors, Section 5.0 [see *J. Chem. Research (S)*, 1999, Issue 1]; there is therefore no corresponding material in *J. Chem. Research (M)*.

View Article Online


off, and washed with water. The solid thus obtained was recrystallized from a 50% acetone aqueous solution or benzene. Mp 93–94 °C.

Tetrameric acetone peroxide was identified by molecular weight determination, elemental analysis, FTIR (KBr), $^{1}H$ and $^{13}C$ NMR and MS. A molecular weight determination was performed on a Knauer Vapour Pressure OSMOmeter (Germany) and the molecular weight was 290 (calc. 296). Elemental analysis was performed on an Elementar Vario EL instrument (Germany). (Found: C, 48.76; H, 8.25. $C_{12}H_{24}O_8$ requires C, 48.64; H, 8.16%). The above analysis was repeated three times. A strong broad band for the carbon–oxygen stretching frequency is seen around 1195 $cm^{-1}$ in the FTIR spectrum. NMR spectra were obtained on a Bruker MSL-90 NMR spectrometer (Germany). $^{1}H$ NMR shows only a methyl signal ($\delta$ 1.43, $CDCl_3$ solvent, TMS as internal standard). $^{13}C$ NMR determination shows two signals with the methyl carbon and carbon bound to the oxygen appearing at $\delta$ 21.3 and 107.4, respectively, which is analogous to trimeric acetone peroxide.[12] The major peaks in the mass spectrum are at *m/z* 74, 58 (base peak) and 43. The molecular ion is not observed in the mass spectrum, indicating that it is unstable.

We thank the Liaoning Natural Science Foundation of China for financial support.

*Received, 22nd December 1998; Accepted, 20th January 1999*
*Paper E/8/09955C*

## References

1 R. A. Sheldon and J. K. Kochi, *Metal-Catalyzed Oxidation of Organic Compounds,* Academics Press, New York, 1981.
2 G. Strukul, *Metal Catalysis in Hydrogen Peroxide Oxidations,* Reidel, Dordrecht, The Netherlands, 1991.
3 P. Battioni, J. P. Renaud, J. F. Bartoli and K. Mansuy, *J. Chem. Soc., Chem. Commun.*, 1986, 341.
4 J. Muzart and N'Ait Ajjou, *J. Mol. Catal.*, 1991, **66**, 155.
5 H. H. R. Barton, P. E. Eaton and W. G. Liu, *Tetrahedron Lett.*, 1991, **32**, 6263.
6 K. Paul, P. R. Story, P. Busch and J. R. Sanderson, *J. Org. Chem.*, 1976, **41**, 1283.
9 P. R. Story, B. Lee, C. E. Bishop, D. Denson and P. Bush, *J. Org. Chem.*, 1970, **35**, 3059.
8 J. R. Sanderson, R. J. Wilterdink and A. G. Zeiler, *Synthesis*, 1976, 479.
9 A. V. Tobolsky and R. B. Mesrobian, *Organic Peroxides, Their Chemistry, Decomposition, and Role in Polymerization,* Interscience, New York, 1954, p. 48.
10 M. S. Kharasch and M. Gladstone, *J. Chem. Educ.*, 1939, **16**, 498.
11 G. Rothenberg, H. Wiener and Y. Sasson, *J. Mol. Catal. A: Chem.*, 1998, **136**, 253.
12 *Carbon-13 NMR,* Sadtler Research Laboratories. Inc, 1978, 4610C.

Published on 01 January 1999 on http://pubs.rsc.org | doi:10.1039/A809955C
Downloaded on 17 March 2013